

Exploring uncertainty in Antarctica's future sea level contribution

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Total word count: 4566 words.

Chapter 1

Literature Review

1.1 Introduction

615 words

The IPCC's Sixth Assessment Report (AR6) identified sea level rise as “an existential threat for island nations, low-lying coastal zones, and the communities, infrastructure, and cultural heritage therein” (Cooley et al., 2022). A rise of 1 m would increase the global population living below mean sea level from between 34–49 million people today to 77–132 million (Seeger and Minderhoud 2026, with uncertainty due to the choice of digital elevation model and population dataset). This figure also belies the variability in local sea level over multiple timescales (Woodworth et al., 2019) — storm surges and other short-timescale variability are a hazard to coastal communities living in low-elevation zones above mean sea level. Although this thesis concerns itself only with long-timescale sea level change, it is important to understand that the impacts of sea level rise are not felt in a binary way (i.e. land lies either above or below sea level), but rather as a continuously increasing risk to which coastal commu-

nities are exposed (Boumis et al., 2023). This exposure is highly regional, with vulnerability concentrated in the Global South (Hooijer and Vernimmen, 2021; Rentschler et al., 2023; Monioudi et al., 2025). Consequently, the communities most exposed to sea level rise are often also the least economically and institutionally equipped to adapt.

NOTE: I could say something here about local sea level change, vertical land movement, gravitational effects, etc.

Since the early 20th century, global mean sea level rise has accelerated from $1.35 \pm 0.57 \text{ mm yr}^{-1}$ between 1901–1990 to $3.69 \pm 0.48 \text{ mm yr}^{-1}$ in 2006–2018 (Fox-Kemper et al., 2021). AR6 assesses with high confidence that this acceleration is the result of anthropogenic warming and that future sea level rise is sensitive to future greenhouse gas emissions. It projects a likely (66% chance) sea level rise of 0.33–0.62 m by 2100 under a low emissions pathway and 0.63–1.01 under a high emissions pathway, the upper bound of which coincides with the 1 m sea level rise hypothetical evaluated by Seeger and Minderhoud (2026). Sea level rise occurs either through thermosteric effects (“thermal expansion”) or through barystatic contributions (i.e. Mass flux) from another source such as land water storage, mountain glaciers, or the Greenland and Antarctic ice sheets. The dominant drivers of historic sea level rise have been thermal expansion and mountain glaciers. However, contributions from the ice sheets have accelerated since the 1970s and are projected to be significant sources of sea level rise this century (Fox-Kemper et al., 2021). **NOTE: I’m uncertain about the etiquette for citing the same report multiple times in one paragraph. Have I done this appropriately?**

Beyond 2100, AR6 cites “deep uncertainty” in multi-century sea level projections, due in large part to poorly constrained processes governing the Antarctic Ice Sheet response (Edwards et al., 2021). The 21st century projections in AR6

are informed heavily by the Ice Sheet Model Intercomparison Project (Nowicki et al., 2016). However, no such project existed for 2300 projections at the time. Instead, AR6 relied on a combination of methods, including extrapolation of 2100 rates of mass loss and the expert elicitation by Bamber et al. (2019). They also considered highly uncertain processes related to the failure of tall ice cliffs, which was only included in a handful of modelling studies at the time, but which dramatically increase Antarctica’s mass loss on these timescales (DeConto and Pollard, 2016). Even excluding this process, AR6 projected 0.3–3.1 m of sea level rise by 2300 under a low emissions pathway and 1.7–6.8 m under high emissions. Synthesized estimates for the individual sources were not provided, but uncertainty in Antarctica’s contribution was wider than 3 m in some assessments of the high emission pathway. Constraining this uncertainty is therefore the main obstacle to more confident multi-century sea level projections. Since AR6, an increased number of studies have simulated the Antarctic response to multi-century climate change (see Section 1.5.1), but identifying and constraining the uncertainty in its sea level contribution remains challenging, thus demonstrating the need for further work.

1.2 The Antarctic Ice Sheet

1.2.1 Geographical setting

485 words

The Antarctic ice sheet is the largest store of freshwater on Earth, holding 26.0 ± 0.4 million km³ of ice (Morlighem et al., 2020). It can be split into three climatically distinct regions: West Antarctica, East Antarctica, and the Antarctic Peninsula. The West Antarctic Ice Sheet (WAIS) is grounded on bedrock that lies predominantly below sea level (a marine ice sheet), which makes it

susceptible to instabilities that could bring about self-sustaining mass loss (see Section 1.3.6). This is particularly concerning because West Antarctica is the region with the highest current rate of mass loss (Table 1.1) and may have collapsed completely in past warm climates (see Section 1.2.3). West Antarctic glaciers drain into the Amundsen Sea, Ross Sea, and Weddell Sea sectors of the Southern Ocean. The latter two feature Antarctica's two largest floating ice shelves, the Ross ice shelf (**TODO: value** km²) and the Filchner-Ronne ice shelf (**TODO: value** km²), respectively. The ocean cavities beneath them are key regions for the production of dense deep waters that form part of the global ocean circulation (Rintoul et al., 2025).

Table 1.1: Antarctica's component ice sheets, their rate of mass loss between 1992–2020 (Otosaka et al., 2023), and their total sea level potential (Morlighem et al., 2020).

Region	Mass loss (Gt yr ⁻¹)	Sea level potential (m)
WAIS	-82 ± 9	5.3 ± 0.2
EAIS	3 ± 15	52.2 ± 0.7
APIS	-13 ± 5	0.27 ± 0.02

The East Antarctic Ice Sheet (EAIS) is separated from West Antarctica by the Trans-Antarctic mountains, a steep mountain range with over 4000 m peaks. It represents by far the largest share of Antarctica's sea level potential, but has contributed the least to recent sea level change (Table 1.1). For this reason and others, the EAIS has historically been assumed to be more stable to climate change than other parts of Antarctica, but more recent evidence brings this assumption into question (Stokes et al. 2022 and references therein). Although the EAIS is partly terrestrial (grounded above sea level), it has three key marine basins: The Recovery, Wilkes, and Aurora subglacial basins. Each of these holds ice with a sea level potential comparable to the

West Antarctic Ice Sheet (Morlighem et al., 2020). **NOTE: Removed section on Amery ice shelf. I've tried to be a bit more minimal here, but it now feels quite bare with basically just the Stokes reference. Have I gone too far?**

The Antarctic Peninsula Ice Sheet (APIS) is a comparatively small part of Antarctica, but is losing mass faster than any other region relative to its size (Table 1.1). It is distinct as the most Northerly region of the continent, reaching 63°S towards the Drake Passage. Its mountain range features smaller, Alpine-like glaciers down steep terrain. The Peninsula acts as an orographic barrier between the Weddell Sea and the Bellingshausen Sea, strongly influencing the climate on either side (Davies et al., 2026). The largest ice shelf in the APIS, the Larsen ice shelf, extends down the Eastern flank of the peninsula and is divided into smaller sections Larsen A to D, the largest of which is Larsen C at **TODO: value and cite** km². The recent changes in the Antarctic Peninsula are significant not only to the local environment, but also serve as a bellwether for processes that may later occur in East and West Antarctica with substantially greater consequences for sea level.

1.2.2 Recent observed changes

Antarctica is not in equilibrium with the current climate, with some areas showing a clear trend in mass balance that is consistent across multiple methods of measurement (e.g. Gardner et al. 2018; Smith et al. 2020; Velicogna et al. 2020). Furthermore, mass loss from West Antarctica has accelerated in every four year period from -37 ± 19 Gt yr⁻¹ in 1992–1996 to -131 ± 21 Gt yr⁻¹ in 2012–2016, before slowing to -94 ± 25 Gt yr⁻¹ in 2017–2020 (Otosaka et al., 2023). The vast majority of this mass loss has originated from the Amundsen Sea Sector of West Antarctica, particularly the Thwaites and Pine Island

Glaciers (Rignot et al., 2019). Sediment records from the Amundsen Sea indicate that Thwaites and Pine Island Glaciers have been in disequilibrium since at least the 1940s **TODO: cite**. Direct measurements of Pine Island Glacier made possible by the satellite era reveal a consistent pattern of change: retreat of the glacier terminus, acceleration of its ice stream, and a subsequent thinning as the glacier adjusts to maintain its flux balance (Joughin et al., 2003; Payne et al., 2004) **TODO: cite more recent**. It is difficult to draw direct links between the current disequilibrium and anthropogenic climate change, but it's likely that climate change has at least exacerbated the retreat of Pine Island glacier over this period (Bradley et al., 2025). **TODO: See Ines's 2023 remote sensing review paper for more inspiration on Pine Island.**

TODO: Thwaites

Unlike the Arctic, which has warmed by **TODO: value** times the global average due to nonlinear feedbacks (Previdi et al., 2021), Antarctic surface air temperatures have broadly followed the global trend **TODO: cite**. The one exception to this is the Antarctic Peninsula, which has warmed three times faster than the rest of the continent, up to temperatures otherwise unprecedented in the Holocene (Vaughan et al., 2003). This atmospheric warming has increased surface melting of the Antarctic Peninsula Ice Sheet and its resulting sea level contribution (Vaughan, 2006). The most pronounced changes in the APIS have been the thinning, retreat, and collapse of some of the peninsula's ice shelves (Cook and Vaughan, 2010; Rignot et al., 2013). The extent of ice shelves on the peninsula roughly trace the -9°C isotherm, suggesting a "limit of viability" for APIS ice shelves that will shrink with further warming (Morris and Vaughan, 2013). Although this is a useful heuristic, the mechanisms of ice shelf collapse are not purely thermal (see, for example, Sections 1.3.3 and 1.3.4).

The first ice shelf collapse to occur during the satellite era was the Wordie

Ice Shelf in 1989 (Doake and Vaughan, 1991). This was followed by a sequential collapse from North to South down the Eastern flank of the peninsula, starting with the Larsen inlet — also in 1989 — then the Prince Gustav and Larsen A ice shelves in 1995 (Cook and Vaughan, 2010). Finally, the Larsen B ice shelf progressively thinned between 1992 and 2001 before its eventual collapse over a few days in February 2002 (Shepherd et al., 2003). The Larsen C ice shelf has so far avoided collapse, but has continued to thin at a rate of -3.8 m per decade over increasingly large areas of the ice shelf (Paolo et al., 2015). Between August 2015 and July 2017, a 200 km crack developed along the ice shelf, leading to the calving of a large iceberg representing 10% of the total ice shelf area (Hogg and Gudmundsson, 2017). Due to the loss of buttressing force provided by ice shelves (see Section 1.3.3), the collapse of each ice shelf on the peninsula was consistently followed by an acceleration and thinning of its upstream glaciers (Rignot et al., 2004; Glasser et al., 2011; Dømggaard et al., 2025). The collapse of the Prince Gustav and Larsen A ice shelves cannot be uniquely attributed to anthropogenic climate change, since sediment records indicate that they have previously collapsed during the mid-Holocene (Pudsey and Evans, 2001; Brachfeld et al., 2003). This is corroborated by written accounts of the Prince Gustav ice shelf by 19th century polar explorers suggesting that it had been in retreat since at least 1843 (Cooper, 1997). However, similar records from ocean sediments previously beneath the Larsen B ice shelf indicate that it had been stable throughout the Holocene, and therefore suggest that its recent collapse contains an anthropogenic signal (Domack et al., 2005).

NOTE: Too much on the Peninsula? Two big paragraphs on the smallest region (and least important for my work) seems like a lot. On the other hand, the peninsula is an excellent case study for what might hap-

pen elsewhere in Antarctica, an argument which I plan to develop in later sections, so it is perhaps worth going into detail here.

In contrast to the WAIS and APIS, sea level contribution from the East Antarctic Ice Sheet remains close to a state of balance, with a slight positive mass trend of $3 \pm 15 \text{ Gt yr}^{-1}$ between 1992–2020 (Otosaka et al., 2023). The high uncertainty in this estimate is the result of disagreement between different methods of measurement, with some studies showing a negative mass trend over similar periods (e.g. Rignot et al. 2019; Shepherd et al. 2019). These aggregate measurements also bely the spatial variability in East Antarctic mass trends. Several studies highlight regional mass loss from Wilkes Land in East Antarctica, particularly the Totten Glacier, which drains the Aurora subglacial basin (Shepherd et al., 2019; Velicogna et al., 2020; Smith et al., 2020). This is paired with a positive mass trend over the East Antarctic plateau, consistent with reanalysis-derived increases in precipitation (Davis et al., 2005). This spatial variability suggest that whilst the East Antarctic Ice Sheet is in a state of balance, it is not in a state of equilibrium, with the potential for more significant mass change if this balance is disrupted.

TODO: Check Jamie's and Mira's theses to look for more EAIS notes.

TODO: Quick synthesis of this section?

1.2.3 Antarctica in past climates

Due to the long timescales involved with ice sheet evolution, and the limited time with which satellites have been able to measure them, models are limited in their capacity to reduce uncertainty through comparison with modern observations. This motivates the use of past climates as a possible route to constrain model behaviour (Tierney et al., 2020). Past climates are particularly useful for model calibration because they can be so different to the present, testing the

model's ability to capture larger changes in the ice sheets than we observe in the modern era.

Palaeoclimate reconstructions reveal that the Mid-Pliocene Warm Period (mPWP), with CO₂ concentrations of 360-420 ppm and temperatures 2.5-4 °C warmer than present (Fox-Kemper et al., 2021), is the closest analogue to end of the century projections under medium to high emissions (Burke et al., 2018). Model experiments of Antarctica show that temperatures during the Pliocene were high enough to collapse the WAIS completely during interglacials and regrow it again during glacial periods (Pollard and DeConto, 2009). However, these results are very model-dependent, with a larger ensemble producing a broad range of Pliocene sea levels between 6-45m higher than present (Dutton et al., 2015). The use of climate proxies such as benthic foraminiferal $\delta^{18}\text{O}$ values or grain size produce a more tightly constrained range of 12-25 m (Miller et al., 2012; Grant et al., 2019). These estimates are limited to a single site and cannot exactly represent global mean sea level, but do give some indication that at least partial collapse of the WAIS occurred during the mPWP. Proxies more directly taken from the Ross and Amundsen seas, which neighbour the WAIS, appear to corroborate this with evidence of growth/retreat over glacial-interglacial cycles (Naish et al., 2009; Gohl et al., 2021). This combined evidence from models and proxies create a strong argument that the WAIS collapsed fully during the Pliocene and may be vulnerable to collapse in future warm climates.

Given the slight evidence of instability in marine sectors of the EAIS in the modern era, there is also significant interest as to whether the EAIS remained stable during the mid-Pliocene. The higher estimates of global mean sea level during the Pliocene would necessitate at least some contribution from the EAIS, but the lower estimates could be met with ice loss only in Greenland and West

Antarctica (Grant et al., 2019). The best evidence for EAIS retreat therefore comes from proxies located directly in East Antarctica, though this is a region that is extremely difficult to access and data is therefore limited. Marine sediment cores taken offshore from the Wilkes subglacial basin indicate that the region used to feature much lower coverage of sea ice, with the polar front pushed further inland towards the glacier terminus (Cook et al., 2013; Bertram et al., 2018). Ice rafted debris outside of Prydz Bay also suggests that the EAIS discharged large “armadas” of icebergs as it responded to high temperatures in the Miocene and Early Pliocene (Williams et al., 2010), and may have remained in a reduced state until as late as the Pleistocene (Wilson et al., 2018). There are comparatively fewer studies that estimate ice volume in the Recovery or Aurora basins during the Pliocene, although the evidence that does exist points to some retreat (Aitken et al., 2016).

1.3 Ice sheet dynamics

1.3.1 Ice rheology*

NOTE: I may want a section here to talk about the polycrystalline structure of ice, as this is relevant to the flow law and the uncertain value of Glen’s exponent, but perhaps this is getting too into the weeds.

1.3.2 Ice-atmosphere interactions*

NOTE: Not particularly relevant to my thesis given that we take SMB directly from GCMs. However, a quick review of lapse rates might be useful when discussing why we leave them out later. Anything else in ice-atmosphere that would be good to talk about?

1.3.3 Ice-ocean interactions

415 words

The Southern Ocean is a confluence of water masses transported through the global ocean circulation. Deep water from the Pacific, Indian, and North Atlantic Oceans mix to form Circumpolar Deep Water (CDW), which circulates through the Antarctic Circumpolar Current (ACC) at depths of up to 3000 m (Talley, 2013). CDW is characterised generally by much warmer temperatures and higher salinity than the surface waters above it (Wang et al., 2025). It can be divided into 'upper' and 'lower' water masses with distinct characteristics. Upper CDW includes contributions from Indian and Pacific deep waters, which are warm, oxygen-poor, and nutrient-rich; Lower CDW contains more North Atlantic Deep Water, giving it characteristically high salinity (Whitworth et al., 2013). Upper CDW shoals as it moves Southwards, eventually mixing with surface waters at the Southern boundary of the ACC. However, lower CDW is dense enough to remain below the mixed layer and extend further towards the Antarctic continent (Orsi et al., 1995). The melting of Antarctic ice shelves is predominantly a question of whether this CDW is able to access the continental shelf that surrounds Antarctica (Wang et al., 2023).

Access to the shelf is limited by the Antarctic Slope Front (ASF), a density front along the shelf break, and a corresponding westward Antarctic Slope Current (ASC) (Jacobs, 1991). The ASF acts as a barrier between CDW and the cooler, fresher shelf waters that occupy ice shelf cavities around Antarctica. The structure of the ASF is highly variable across different sectors of Antarctica and is sensitive to local wind stresses (Spence et al., 2014) and freshwater fluxes (Si et al., 2023). These local conditions give rise to three distinct ice shelf cavity regimes: cold and fresh cavities, cold and saline cavities, and warm and saline cavities.

In Warm and saline cavities such as those in the Amundsen Sea, a weak ASF allows CDW to encroach onto the shelf through

The ASF is weakest in the Amundsen Sea, where the Eastward Antarctic Circumpolar Current dominates over the westward ASC (). This Eastward flow is associated with a polewards Ekman transport in the layer below, which helps to transport CDW onto the continental shelf (Klinck and Dinniman, 2010). This oc

The Southern boundary of the ACC flows very close to the continental shelf in the Amundsen and Bellingshausen Seas, but is deflected away from the Antarctic continent by cyclonic gyres in the Weddell and Ross Seas (Orsi et al., 1995). Temperatures in the gyres are generally lower than those found in ACC (Wang et al., 2023), but some heat is transported to the shelf break along the Eastern boundary of each gyre via entrained CDW (Assmann and Timmermann, 2005; Ryan et al., 2016).

have strengthened and shifted polewards as a response to greenhouse gas emissions and ozone depletion (Vaughan et al., 2013; Lee and Feldstein, 2013).

1.3.4 Calving and hydrofracture*

NOTE: Placemaker section in case I have time to discuss this in more detail.

1.3.5 Subglacial hydrology*

NOTE: Placemaker section in case I have time to discuss this in more detail.

1.3.6 Ice sheet instability

Since the 1970s, it has been suggested that the West Antarctic ice sheet may undergo rapid deglaciation in a warming climate (Mercer, 1978). This has evolved into the theory of Marine Ice Sheet Instability (MISI), which states that the grounding line position of a marine ice sheet on a retrograde slope (i.e. the bed becomes deeper towards the centre) is inherently unstable (Schoof, 2007). The fundamental principle of MISI is that discharge at a glacier's terminus is proportional to the glacier thickness at its grounding line. The retreat of the grounding line down a retrograde slope increases thickness, which in turn increases discharge, which causes the grounding line to retreat further, creating a feedback loop. In practice, this effect is mitigated by ice shelves, which buttress the ice sheet, generating sufficient friction to stabilise the grounding line (Joughin and Alley, 2011). However, Antarctica's ice shelves are also thinning as a result of climate change. In 2002 the Larsen A and B ice shelves on the Antarctic peninsula collapsed over just a few days (Shepherd et al., 2003).

The West Antarctic Ice Sheet is a strong candidate for undergoing MISI, since it overlies a deep bathymetric basin up to 2000m below sea level, giving it an overall retrograde slope. This is supported by satellite measurements of Antarctic mass balance, which identify the Amundsen Sea Embayment in West Antarctica as the primary region for ice loss in recent decades (Smith et al., 2020; Velicogna et al., 2020; Otosaka et al., 2023). Pine Island glacier, which terminates in the Amundsen Sea, shows evidence of Circumpolar Deep Water at its grounding line, retreat down a retrograde slope, thinning of its ice shelves, and an acceleration in its discharge (Joughin et al., 2003; Jenkins et al., 2010; Jacobs et al., 2011; Favier et al., 2014). It has been proposed that West Antarctica could be approaching a 'tipping point', in which MISI could push the ice sheet into a dramatically new state with only a small change in

forcing (Pattyn and Morlighem, 2020; Rosier et al., 2021; Armstrong McKay et al., 2022). Regional ocean models that are able to resolve the temperature in ice shelf cavities find that further warming is unavoidable under any future scenario (Naughten et al., 2023), indicating that collapse of the West Antarctic ice sheet may now be inevitable.

Compared to the WAIS, the EAIS has been broadly considered to be relatively stable. However, the EAIS, whilst mostly above sea level, does include some marine sectors which may be vulnerable to MISI (Armstrong McKay et al., 2022). As with the Amundsen Sea Embayment, warm Circumpolar Deep Water has been observed infiltrating the ice shelf cavities and driving thinning of the Totten glacier, which drains part of the Aurora subglacial basin (Greenbaum et al., 2015; Aitken et al., 2016; Rintoul et al., 2016; Silvano et al., 2017). The ice shelf at the terminus of the Cook glacier, which drains a large proportion of the Wilkes basin, collapsed between 1973-1989, and basal thawing has recently been observed along the nearby Adélie-George V coast (Miles et al., 2018; Dawson et al., 2024). However, warming over the Southern ocean also drives increased precipitation over East Antarctica (Nicola et al., 2023), which has the potential to offset some or all of the mass loss in the marine sectors. Uncertainty in the overall mass balance is large, but a synthesis of multiple modelling and observation studies suggests that it will remain approximately in balance this century, but has the potential to cause multiple metres of sea level rise on multi-century timescales (Stokes et al., 2022).

Other than marine ice sheet instability, additional mechanisms have been proposed that could drive rapid deglaciation in marine segments of Antarctica. Marine Ice Cliff Instability (MICI; Bassis and Walker 2012) is the theory that marine-terminating ice cliffs could be liable to rapid retreat under certain conditions. Models that include MICI yield significantly greater sea level rise than

those without (DeConto and Pollard, 2016; DeConto et al., 2021). However, the inclusion of visco-elastic effects in such models indicate that ice cliffs could be more stable than previously thought (Clerc et al., 2019; Bassis et al., 2021). Bathymetric grooves in the ocean floor could hint at evidence of MICI (Wise et al., 2017), but it has yet to be observed in the modern era, nor is it necessary to achieve the sea level change seen in past climates (Edwards et al., 2019). Similarly, corrugation ridges along the ocean floor suggest a 'buoyancy-driven' mechanism for ice sheet retreat, up to hundreds of metres per day along low-gradient slopes (Batchelor et al., 2023). However, like MICI, it has yet to have been observed in the modern climate.

1.3.7 Solid Earth effects

On short to medium timescales, ice-ocean interactions and ice sheet stability are the main dynamics to consider regarding future sea level rise. However, on longer timescales spanning multiple centuries or millennia, additional effects relating to the interaction between the ice sheets, the solid Earth, and its gravitational field, become relevant. These effects are collectively referred to as Glacial Isostatic Adjustment (GIA; Whitehouse 2018, and references therein). As an ice sheet retreats, the load that it exerts on the Earth's crust decreases, inducing a visco-elastic response which uplifts the land surface due to upwards pressure in the asthenosphere. GIA is currently observed in regions that experienced deglaciation following the last ice age, recording uplift of over 10 mm a^{-1} in some parts of North America and Europe (Sella et al., 2007; Lidberg et al., 2010). As evidenced by these regions, which have been without an ice sheet for over 8,000 years, the response to glacial unloading is not instantaneous and can be drawn out for millennia.

GIA has the capacity to significantly influence shoreline migration and bedrock

topography over long timescales, which in turn feeds back into the dynamics driving ice sheet growth or retreat. For example, the uplift of bedrock following unloading could limit the accessibility of Circumpolar Deep Water or reverse/intensify the retrograde slope in a region susceptible to Marine Ice Sheet Instability. Larour et al. (2019) find that GIA mitigates 27% of projected sea level rise over 500 years using a model of the Thwaites glacier in which it is included. Similarly, Kachuck et al. (2020) show that GIA could reduce ice mass loss from the Pine Island Glacier by 30% over 150 years. These modelling studies show that GIA can have an impact on even sub-millennial timescales.

1.4 Ice sheet modelling

68 words

Numerical ice sheet models are important tools for Antarctic science, both for understanding the mechanisms behind recent ice sheet change (e.g. Payne et al. 2004; Gudmundsson et al. 2019; Bradley et al. 2025), exploring possible ice sheet configurations in the deep past (e.g. De Boer et al. 2015) and for making projections of future ice sheet change (Seroussi et al. 2020, 2024, among many others). This section reviews the construction of a general ice sheet model and the important choices involved in simulating the Antarctic ice sheet.

1.4.1 Stokes flow

200 words

Practical ice flow modelling is done using a hierarchy of approximations to incompressible Stokes flow, which is itself the low Reynolds number limit of the Navier-Stokes equations. In this limit, inertial terms are ignored, and flow is

governed by a balance between pressure gradients, viscous stresses, and gravitational driving stresses. These are solved alongside a constitutive relation, which defines the ice rheology (see Section 1.4.2). It is also necessary to impose boundary conditions at the base and surface of the ice sheet. For a rigid boundary (the interface at the bed), this might take the form:

$$\mathbf{u}_b = F(\boldsymbol{\tau}_b), \quad \mathbf{u} \cdot \mathbf{n} = -m_b, \quad (1.1)$$

where $\mathbf{u}$ is the ice velocity field, $\mathbf{u}_b$ is the velocity component parallel to the bed, $\mathbf{n}$ is the normal to the bed, $\boldsymbol{\tau}_b$ is the basal shear stress, F is a sliding law (see Section 1.4.4), and m_b is the basal melt rate (in velocity units). For a free surface (the interface with the atmosphere or ocean), dynamic and kinematic boundary conditions are applied to specify the boundary stresses and surface evolution, respectively. Some model setups include an advection-diffusion equation for temperature, which couples to the momentum (Stokes) equations through the constitutive relation (Eq. 1.2). However, assuming isothermal ice (i.e. fixed temperature) is often a reasonable approximation over short timescales.

1.4.2 Constitutive relation

227 words

The constitutive relation describes the relationship between the stress and deformation of a material. For ice, it is common to use Glen's flow law (Glen, 1955):

$$\dot{\epsilon}_{ij} = A\tau^{n-1}\tau_{ij}, \quad \tau = \sqrt{\frac{1}{2}\tau_{ij}\tau_{ij}}, \quad (1.2)$$

where ϵ_{ij} is the strain rate tensor, τ_{ij} is the deviatoric stress tensor, $A(T)$ is a temperature-dependant rate factor, and n is a scalar exponent ("Glen's expo-

nent”). Early work to estimate the value of n for polycrystalline ice suggested that $n \approx 3$ (Weertman 1983 and references therein). Subsequently, $n = 3$ became standard across a wide range of ice sheet models (e.g. Larour et al. 2012; Cornford et al. 2013; Lipscomb et al. 2019). However, Goldsby and Kohlstedt (2001) demonstrated that creep can be divided into multiple regimes, occurring through different mechanisms within the polycrystalline structure. The appropriate value of n depends on the dominant creep mechanism, which in turn depends on the stress. **NOTE: I could talk here about typical driving stresses in Antarctica and the corresponding value $n = 1.8$, but my attempts at this have gotten a bit bogged down and wordy, so have removed for better flow.** Other studies suggest that observations of Antarctic ice streams and ice shelves may be better explained by a higher value of n (e.g. Cuffey and Kavanaugh 2011; Millstein et al. 2022). Ice sheet models that have explored this found Antarctic sea level contribution to be sensitive to the choice of n , with higher values leading to greater sea level rise (Rosier et al., 2025; Getraer and Morlighem, 2025; Martin et al., 2026). **NOTE: Is signposting like “We will explore this uncertainty in chapter 2” (or similar) appropriate in these situations?**

TODO: Cuffey and Paterson and have a good section on this — worth a brief rewriting after reading

1.4.3 Approximations in ice flow models

39 words

Full Stokes modelling is computationally expensive and so is rarely deployed for continent-scale ice sheet modelling (with some notable exceptions, e.g. Seddik et al. 2012). Approximations are typically applied to match the requirements of a particular ice-dynamical regime.

1.4.4 Basal sliding

1.4.5 Surface mass balance*

1.4.6 Sub-shelf melting

1.4.7 Glacial isostatic adjustment

1.5 Projections of the Antarctic ice sheet

1.5.1 Multi-century-scale projections

1.5.2 Uncertainty Quantification

1.6 Uncertainty quantification*

1.6.1 Experiment design

1.6.2 Gaussian Process Emulation

1.6.3 Calibration

Chapter 2

Methods

This is the methods chapter.

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